

Part B: Fill in the Blank (34 questions)

Write the missing word(s). Two prompts per module (items 86–119).

Module 01

86. Science relies on evidence gathered through ___ and experimentation.

87. From molecule to biosphere, levels such as cell and ___ describe organization of life.

Module 02

88. Atoms bond by sharing electrons in ___ bonds.

89. Solutions with more H^+ than pure water are called ___.

Module 03

90. Enzymes are mostly ___ molecules that catalyze cellular reactions.

91. DNA stores genetic instructions using paired ___ bases.

Module 04

92. Plant cells often contain chloroplasts for photosynthesis; animal cells rely heavily on ___ for aerobic ATP.

93. The ___ directs traffic of proteins made on ribosomes.

Module 05

94. Passive transport moves substances ___ concentration gradients without ATP pumps.

95. Cells placed in hypotonic fluid tend to gain water by ___.

Module 06

96. Cells harvest chemical energy from glucose using pathways such as glycolysis and aerobic ____.

97. Coenzymes such as NAD^+ carry electrons during ____ breakdown.

Module 07

98. Transcription builds RNA using a DNA ____ strand.

99. Ribosomes decode ____ during translation.

Module 08

100. During ____, sister chromatids separate toward opposite poles.

101. Crossing over during ____ I exchanges segments between homologous chromosomes.

Module 09

102. Epithelium often forms protective ____ on body surfaces.

103. Bone stores calcium salts and flexible ____ fibers for strength.

Module 10

104. An organism with two different alleles is ____ at that locus.

105. Sex-linked traits involve genes carried on ____ chromosomes.

Module 11

106. Bone-removing cells called ____ degrade mineral matrix during remodeling.

107. Movement at many joints relies on lubricating ____ fluid.

Module 12

108. Contraction cycles require ATP, actin, ____, and calcium ions.

109. Each motor unit pairs one motor neuron with its muscle ____.

Module 13

110. Voltage-gated sodium channels help initiate action ____ along axons.

111. Neurotransmitters diffuse across the synaptic ____ after vesicle release.

Module 14

112. Gram-negative bacteria often have thinner peptidoglycan and an outer ____.

113. Horizontal gene transfer can spread antibiotic resistance via bacterial ____.

Module 15

114. The left ventricle pumps oxygen-rich blood into the ____.

115. Blood picks up most oxygen in pulmonary ____ surrounding alveoli.

Module 16

116. Food chains link producers to successive ____ levels.

117. Populations cannot grow forever because environments have finite ____.

Module 17

118. Natural selection acts on heritable ____ already present in populations.

119. Reproductive ____ separates populations on the path to new species.

Part C: Short Answer / Free Response (34 questions)

Answer clearly in complete sentences where appropriate (items 120–153). Two prompts per module.

120. Describe one way observation differs from opinion in biology lab work.

121. Explain why maintaining stable internal conditions (homeostasis) helps cells function.

122. Contrast ionic and covalent bonding using water or salt as an example.

123. Explain how pH relates to hydrogen ion concentration in everyday terms.

124. Relate protein shape to its ability to act as an enzyme.

125. Describe why nucleic acids store information better than simple sugars alone.

126. Compare roles of mitochondria and chloroplasts as energy-related organelles.

127. Explain why ribosomes exist in both prokaryotic and eukaryotic cells.

128. Differentiate passive transport from active transport with one example each.

129. Explain what happens to a red blood cell placed in very salty water.

130. Outline why cells need ATP even when resting.

131. Contrast aerobic respiration with fermentation in terms of oxygen use.

132. Trace information flow from DNA to functional protein in three short steps.

133. Define mutation and give one reason mutations matter for evolution.

134. Explain why mitosis preserves chromosome number while meiosis halves it.

135. Describe crossing over and state why it increases variation.

136. Give one example each of epithelial, connective, and muscular tissue roles.

137. Explain why cartilage differs from bone when resisting compression.

138. Solve conceptually: Why can carriers appear healthy for recessive disorders?

139. Explain pedigree symbols versus Punnett squares—what each tool shows best.

140. Describe how bones remodel after microdamage during exercise.

141. Explain synovial joint lubrication and why it reduces friction.

142. Explain role of calcium ions in starting muscle contraction.

143. Describe recruitment of motor units as workload rises.

144. Explain resting potential versus depolarization at a neuron membrane.

145. Contrast sympathetic and parasympathetic effects on heart rate.

146. Explain why completing an antibiotic course reduces resistant survivors.

147. Differentiate bacteria from viruses regarding metabolism and replication.

148. Describe pulmonary circulation versus systemic circulation in one paragraph.

149. Relate partial pressures of oxygen/carbon dioxide to gas exchange at alveoli.

150. Explain energy flow versus nutrient cycling using a simple food chain.

151. Define carrying capacity and give one factor that lowers it.

152. Explain natural selection using variation, inheritance, and differential reproduction.

153. Describe one reproductive isolating mechanism that can lead to speciation.

End of Practice Test 12